**VA MEDICAL CENTER**
**Atlanta, Georgia**

**July 2024**

**ENGINEERING SERVICE**
**NUMBER 138-9-2024**

# HAZARDOUS ENERGY CONTROL PROGRAM (LOCKOUT/TAGOUT)

1. **PURPOSE**: To prevent serious or fatal injuries by establishing minimum requirements for lockout or energy isolating devices on machinery or equipment, whenever maintenance or service is done on them where the unexpected energizing, startup or release of stored energy could cause injury.

2. **POLICY**: Engineering Service establishes a program consisting of energy control procedures, employee training and periodic inspections to ensure that before any employee performs any servicing or maintenance on a machine or equipment where the unexpected energizing, startup or release of stored energy could occur and cause injury, the machine or equipment shall be isolated from the energy source, and rendered inoperative. Attachment A contains definitions of the three types of trained employees (authorized, affected, and other); general definitions, job titles of employees authorized and required to lockout machinery or equipment; employees affected by tagout conditions; and other employees exposed to tagout conditions in their work areas.

3. **RESPONSIBILITIES**:

a. The Chief, Engineering Service, is responsible for evaluating and updating the Hazardous Energy Control Program; ensuring that the policy is being followed by supervisors and maintenance/service personnel; and providing on the spot corrections when unsafe work practices are cited.

b. The Associate Chief, Utilities and Plant Operations, is the designated representative to oversee the implementation and management of the LOTO policy and program for Atlanta VAMC. The Facility Operations Supervisors are the designated representative to oversee the implementation and management of the LOTO policy and program for their respective facilities (Ft. McPherson and Carrollton).

c. Supervisors are responsible for certifying that employee training has been accomplished, and the importance of lockout/tagout procedures is understood to ensure full compliance with this policy. Supervisors shall audit employees' compliance with this policy at least annually, using Attachment B (Annual Energy Control Inspection Report). Supervisors shall provide initial training to each new or transferred authorized maintenance employee, and retraining as necessary, to cover the purpose and use of the lockout procedures. The material in Attachment E is to be used by the supervisors for Hazardous Stored Energy Training.

d. Authorized maintenance employees are responsible for locking out and/or tagging out any switch, control device or valve on all equipment to be shut down for maintenance or service.

e. Whenever outside servicing personnel (contractors, etc.) are to be engaged in activities covered by this memorandum, the Engineering Service representative and the outside employer shall inform each other of their respective lockout/tagout procedures. The Engineering Service representative shall ensure that Engineering Service employees understand and comply with the restrictions and prohibitions of the outside employer's energy control program.

4. **PROCEDURES**:

a. The following categories describe three types of equipment in which unexpected energization or startup, or release of stored/residual energy could cause injury:

(1) Cord and plug-connected, no stored energy, single energy source (examples: vacuum cleaner, electric drill, and toaster).

(2) No cord or plug-connected, no stored energy after shutdown (example: An air handler unit).

(3) Potential stored energy, re-accumulation of stored energy, possible multiple energy sources, lockout cannot be achieved and employee workplace exposure to tagout conditions exists, non-exclusive control by maintenance employee (examples: elevators, laundry equipment, and transformers).

**CAUTION**: Employees authorized to perform lockout/tagout procedures shall be certain which switch (es), valve(s), or other isolating devices apply to the equipment to be locked out. Particular attention must be paid to stored potential energy (such as power press machines, which could fall). More than one energy source (electrica1, mechanica1, hydraulic, or others) may be involved. **If more than one energy source is involved, or if any of the conditions in Category 3 (above) apply, there must be machine-specific procedures posted at the machinery/equipment, or at the main electrical disconnect.** Attachment C (Lockout/Tagout Decision Chart) shall be used to determine whether machine-specific procedures are required. Attachment D shall be used to develop machinery/equipment-specific procedures.

b. Electrical or stored energy sources, switch control devices, or valves will be locked out and/or tagged out with the date and name of the individual opening the circuit (or closing the valve) appearing on the tag. Tags shall be durable enough to withstand the environment to which they are exposed, and shall provide a warning statement such as "Do Not Start," or "Danger ---Do Not Energize," etc. Tags must be securely attached so they cannot be inadvertently or accidentally detached.

c. Any device so tagged will be returned to its normal position only by the person whose name is on the tag or, in an emergency, his/her Supervisor. Should a crew change be necessary, the next person providing maintenance shall install his or her own lock and/or tag.

d. Machinery without lockout capabilities must have required specific procedures for energy control. Attachment D is to be used for wiring specific procedures involving this type of machinery.

e. In the event that the individual whose name appears on the lockout or tagout is incapacitated or unavailable to remove the device, that individual's supervisor will do the following:

(1) First - Make every attempt to contact and inform the individual that the lockout/tagout is being moved.

(2) Ascertain the safety of the circuit or device.

(3) Third - Remove the tag or lock personally, when appropriate.

The equipment/machinery may then be energized and tested, or other de-energized systems installed.

f. Stored energy shall be blocked to prevent the accidental operation of the equipment/machinery to be serviced or repaired.

g. The following generic procedures describe proper shutdown and power restoration on equipment or machinery classified in Categories 1 and 2 (above). Note that specific procedures for proper shutdown and power restoration on equipment or machinery in Category 3 (above) are kept in the immediate vicinity of the equipment/machinery.

(1) Single Electrical Circuits

(a) Procedure to open a breaker:

(1) Identify all equipment which will be affected, using prints, schematics, etc.

(2) Notify any users whose circuits will be affected by the loss of power.

(3) Open the circuit and attach a lockout device and tag.

(4) Operating Circuits must be locked out using a padlock or other suitable lockout device.

(b) Procedures to restore a breaker:

(1) Check to ensure that all circuits are clear.

(2) Remove lockout device as necessary.

(3) Close circuit breaker.

(4) Check to ensure that all affected areas are operating normally.

(5) Remove lockout device and tag.

(2) Multiple Electrical Circuits

Procedure to open breakers:

(a) Identify all equipment which will be affected, using prints, schematics, etc.

(b) Obtain permission to open the breakers from the Utility Management Supervisor, the Associate Chief, Utility and Plant Operations, or respective Facility Operations Supervisor.

(c) Notify any users whose circuits will be affected by the loss of power.

(d) Open the circuits and attach "Danger" tags. Each tag must contain the breaker panel number, list of sub-panels affected, and the signature of the technician.

(e) All Operating Circuits must be locked out using padlocks or other suitable lockout devices.

(3) Water Valves

(a) Procedure to close valves:

(1) Identify all equipment which will be affected, using prints, schematics, etc.

(2) Notify any users whose operation will be affected by the loss of water.

(3) Close valves and attach tagouts or lockouts, if feasible. Identify all valves necessary to control energy.

(b) Procedure to reopen valves

(1) Check to ensure that all lines are clear.

(2) Reopen valves.

(3) Check to ensure that all affected areas are operating normally.

(4) Remove tagouts/lockouts as necessary.

(4) Steam valves

(a) Procedure to close valves:

(1) Identify all equipment that will be affected, using prints, schematics, etc.
(2) Notify any users whose operation will be affected by the loss of steam or steam pressure.
(3) Identify all valves as necessary.
(4) Deplete stored steam pressure, energy and condensate from system.
(5) Verify valves closed and attach locks and tags.
(6) Verify absence of pressure, heat and condensate from system.

(b) Procedure to reopen valves:

(1) Check to insure all lines are clear and system piping is intact.
(2) Remove locks and tags.
(3) Open valves in slow increments.
(4) Check for leaks, check to insure all affected areas are functioning normally.

(5) Hydraulic/Air Systems

(a) Procedure for maintenance to hydraulic or air systems.

(1) Identify all equipment which will be affected, using prints, schematics, etc.

(2) Notify any users whose operation will be affected by the loss of air or hydraulics.

(3) Attach tags. Identify all valves that will be necessary.

(4) Drain air reservoirs and lines prior to disconnecting. Release pressure on hydraulic systems where necessary.

(5) Block any equipment which may fall on or trap personnel who are working on it; i.e., forklift hoist, loading dock levelers, backhoe and dump truck equipment, etc.

(b) Procedure to return equipment to normal operation:

(1) Check to ensure that all tools and equipment are removed from work area.

(2) Remove blocks after work is finished.

(3) Conduct a hydraulic/air system operations check before allowing normal operation.

(4) Remove tags.

(5) Return all equipment to normal and check operation.

h. Supervisors shall certify that employee training has been accomplished and the safety significance of lockout/tagout procedures has been conveyed, to ensure full compliance with the policy. The certification of training shall contain each employee's name and the date of training. Training shall include the information found in Attachment E, as well as the following:

(1) Each authorized employee shall be trained in the recognition of applicable hazards and the means necessary to isolate or control the energy.

(2) Each affected employee shall be instructed in the purpose and use of the energy control procedures.

(3) Contractors or other employees may require training in the elements of the procedures.

i. Inspections:

(1) Periodically (at least annually), inspections will be conducted by supervisors and/or safety personnel to ensure that safe procedures are followed, and that employees are familiar with the requirements of this policy. The evaluations shall be documented using Attachment C and shall be kept on file in the Safety Office and by the respective Supervisor

(2) At any indication that procedures are not followed, or that an employee is not knowledgeable of procedures, employee training shall take place and be re-documented according to paragraph 4.h., on training.

j. Group lockout or tagout:

(1) When servicing and/or maintenance is performed by a crew, craft, department or other group, they shall utilize a procedure which affords the employees a level of protection equivalent to that provided by the implementation of a personal lockout or tagout device.

(2) Primary responsibility is vested in an authorized employee for a set number of employees working under the protection of a group lockout or tagout device (such as an operations lock);

(3) The authorized employee shall ascertain the exposure status of individual group members with regard to the lockout or tagout of the machine or equipment;

(4) When more than one crew, craft, department, etc. is involved, assignment of overall job-associated lockout or tagout control responsibility shall be delegated to an authorized employee designated to coordinate affected work forces and ensure continuity of protection;

(5) Each employee working under the direction of the authorized employee shall affix a personal lockout or tagout device to the group lockout device, group lockbox, or comparable mechanism when he or she begins work, and shall remove those devices when he or she stops working on the machine or equipment being serviced or maintained.

5. **REFERERNCE**: 29 CFR 1910.147.

6. **RESCISSION**: Engineering Service Policy Number 138-9-2015, "Hazardous Energy Control Program (Lockout/Tagout)," dated April 2011.

7. **REVIEW DATE**: July 2027

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Attachments: 5

Distribution: All Engineering Service Employees (138)